Matrices

Part VIII Appendix

Scope

Entries, dimensions, and field assumptions define matrix arithmetic. Complete elimination, rank factorizations, and inverse traces support proofs of equal row and column rank, determinant laws, and Gram positive definiteness. Real, complex, and finite fields require different inner-product conventions.

Prerequisites

Vector spaces, linear maps, and matrix multiplication.

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Appendix route

Learning objectives

- Apply matrix arithmetic and transposition.
- Relate rank, determinant, and inverse.
- State the conditions for symmetric positive definiteness and spectral reasoning.
- D.1 Matrices and matrix operations
- D.2 Basic matrix properties

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Matrices and matrix operations

Unless stated otherwise, dimensions are positive integers, entries are real, and vectors are columns. A rank-zero factorization allows zero intermediate columns and rows, with empty sums defined as 0. A matrix $A \in \mathbb{R}^{m \times n}$ has m rows and n columns. The entry a_{ij} uses the row index first and the column index second. Equality requires equal dimensions and equality of every corresponding entry. Addition and scalar multiplication are defined entrywise.

For example,

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \qquad A^T = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$$

have dimensions 2×3 and 3×2, respectively. Transposition means $(A^T)_{ji} = a_{ij}$. Row-major and column-major storage are memory conventions; the mathematical transpose is defined by this index correspondence.

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Matrices and matrix operations (continued)

The vector $x = (1, -1, 2)^T$ is 3×1, while x^T is 1×3. The standard basis vector e_i has a 1 in position i and zeros elsewhere, with dimension specified by context. Zero matrices also require dimensions. The scalar 0 and a 2×3 zero matrix are different types of object.

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Exact zero positions in common square structures

Type	Entry condition for an $n \times n$ matrix
Diagonal	$i \neq j \Rightarrow a_{ij} = 0$
Identity I_n	Diagonal entries are 1; all others are 0.
Tridiagonal	$\lvert i - j \rvert > 1 \Rightarrow a_{ij} = 0$
Upper triangular	$i > j \Rightarrow a_{ij} = 0$
Lower triangular	$i < j \Rightarrow a_{ij} = 0$
Unit triangular	Triangular, with every diagonal entry equal to 1.
Permutation	Exactly one 1 in each row and each column; all other entries are 0.
Real symmetric	$A = A^T$

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Exact zero positions in common square structures (continued)

These classes can overlap. The identity is diagonal, upper and lower triangular, symmetric, and a permutation matrix. A tridiagonal matrix need not be symmetric because opposite diagonals may differ. Unit triangular does not mean that every entry is 1. A unit vector likewise need not be an all-ones vector.

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Check dimensions and the contracted index

For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, the product $C = AB$ belongs to $\mathbb{R}^{m \times p}$, with

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \qquad 1 \le i \le m, \quad 1 \le j \le p.$$

The shared index k is summed out; the outer indices i, j remain. This is not entrywise multiplication.

Using the earlier A and $B = \begin{bmatrix} 2 & 1 \\ -1 & 3 \\ 4 & 0 \end{bmatrix}$ gives

Entry	Calculation
$(AB)_{11}$	$1 \cdot 2 + 2(-1) + 0 \cdot 4 = 0$
$(AB)_{12}$	$1 \cdot 1 + 2 \cdot 3 + 0 \cdot 0 = 7$
$(AB)_{21}$	$0 \cdot 2 + 1(-1) + 1 \cdot 4 = 3$
$(AB)_{22}$	$0 \cdot 1 + 1 \cdot 3 + 1 \cdot 0 = 3$

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Check dimensions and the contracted index (continued)

Thus $AB = \begin{bmatrix} 0 & 7 \\ 3 & 3 \end{bmatrix}$. The reverse product is also defined, but has size 3×3:

$$BA = \begin{bmatrix} 2 & 5 & 1 \\ -1 & 1 & 3 \\ 4 & 8 & 0 \end{bmatrix}.$$

The products differ even in size. Square products also need not commute; scalar commutativity does not transfer to matrix multiplication.

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Check dimensions and the contracted index (continued)

Check sizes, then sum three aligned k terms

A : 2 rows × 3 columns

1	2	0
0	1	1

×

B : 3 rows × 2 columns

2	1
-1	3
4	0

=

C=AB : 2×2

0	7
3	3

C[1,2] = row1 of A · column2 of B = 1×1+2×3+0×0 = 7

Yellow row/column lengths3 share k. Output retains row1,column2; not entrywise multiplication.

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Implementable multiplication and exact cost

```
MULTIPLY(A[m,n], B[n,p])
    allocate independent C[m,p]
    for i=1..m
        for j=1..p
            s = A[i,1] * B[1,j]
            for k=2..n
                s = s + A[i,k] * B[k,j]
            C[i,j] = s
    return C
```

For $n \geq 1$, this uses mnp scalar multiplications, $mp(n-1)$ additions, and mp output writes, plus loop control. Constant-cost scalar arithmetic in the RAM model gives $\Theta(mnp)$ time, or $\Theta(n^3)$ for square matrices. Initializing $s = 0$ and doing n additions instead adds mp additions. That is a different exact count with the same asymptotic order.

Output entries must not alias input entries that will be read later, since writes could corrupt those reads. Arbitrary-precision integers and field elements require separate bit costs. Performing n^3 scalar multiplications does not guarantee n^3 time for arbitrary bit lengths.

Associativity, distributivity, and compatible dimensions

Let A be $m \times n$, B be $n \times p$, and C be $p \times q$. For every valid output index (i, j),

$$\begin{aligned}((AB)C)_{ij} &= \sum_{k=1}^{p} \sum_{\ell=1}^{n} a_{i\ell} b_{\ell k} c_{kj} \\ &= \sum_{\ell=1}^{n} a_{i\ell} \left(\sum_{k=1}^{p} b_{\ell k} c_{kj} \right) = (A(BC))_{ij}.\end{aligned}$$

Reordering is valid because the sums are finite. Distributivity follows from scalar distributivity at each entry. Identity dimensions must match: $I_m A = A I_n = A$.

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Transposition reverses factor order

For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, compare position (j, i) on both sides of $(AB)^T = B^T A^T$:

$$((AB)^T)_{ji} = \sum_{k=1}^{n} a_{ik} b_{kj},$$

$$(B^T A^T)_{ji} = \sum_{k=1}^{n} (B^T)_{jk} (A^T)_{ki} = \sum_{k=1}^{n} b_{kj} a_{ik}.$$

The entries agree by **scalar** commutativity. The matrix product itself need not commute; transposition reverses the factor order.

For real symmetric matrices A, B of the same size, their sum and difference remain symmetric. Their product is symmetric if and only if $AB = BA$, since $(AB)^T = BA$. Two symmetric inputs alone do not imply a symmetric product.

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Matrix-vector, inner, and outer products

Write the columns of $A \in \mathbb{R}^{m \times n}$ as $a_1, \dots, a_n$. For $x \in \mathbb{R}^n$, the product $Ax = \sum_{j=1}^{n} x_j a_j$ is an m-vector. The earlier example gives

$$Ax = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

which matches the row dot products.

For equal-length real vectors u, v, the inner product $u^T v$ is a scalar and the outer product uv^T is a matrix. With $u = (1, 2)^T$ and $v = (3, -1)^T$,

$$u^T v = 1, \qquad uv^T = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}.$$

Column j is $v_j u$, so the rank is at most one. If both vectors are nonzero, at least one column is a nonzero multiple of u and the rank is one. If either vector is zero, the rank is zero.

The real norm is $\|x\|_2 = \sqrt{x^T x}$. Complex vectors require $x^* x = \sum_i |x_i|^2$, where $*$ denotes conjugate transpose. For $x = [i]$, ordinary transpose gives $x^T x = -1$, which is not squared length.

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Permutations act differently on rows and columns

For

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

one has $Px = (x_3, x_1, x_2)^T$. If A has three rows, PA selects its old rows in the order $3, 1, 2$. In $(PA)_{ij} = \sum_{k=1}^{3} p_{ik} a_{kj}$, each row of P selects one term.

For a matrix M with three columns, MP selects its old columns in the order $2, 3, 1$, because the columns of P are e_2, e_3, e_1. Derive the direction from entries or basis columns, rather than memorizing one order for both sides.

Distinct basis columns give $P^T P = I_3 = P P^T$, so $P^{-1} = P^T$. Products of permutation matrices remain permutation matrices because composing two basis reorderings is a bijection.

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Rank and null vectors

Vectors $v_1, \ldots, v_n$ are linearly independent if $\sum_{j=1}^{n} c_j v_j = 0$ forces every coefficient $c_j = 0$; otherwise they are dependent. Column rank is the largest number of independent columns, and row rank is defined analogously. Zero matrices have rank zero.

Since $Ax = \sum_j x_j a_j$, column independence is equivalent to $Ax = 0$ having only the solution $x = 0$. A nonzero solution is a nonzero null vector. The null space always contains 0; absence of a nonzero null vector does not mean the null space contains no vectors.

The matrix

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}$$

has eight nonzero entries but rank two. Its third column is the sum of the first two, which are independent. A null vector is $(-1, -1, 1)^T$. Rank is neither the nonzero-entry count nor the number of nonzero entries on the original diagonal.

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Elimination operations and intermediate states

Row swaps, nonzero row scaling, and adding a multiple of another row are invertible operations. Each is left multiplication by an invertible elementary matrix. They preserve column-dependence coefficients and row span.

For M, the complete elimination trace is

$$M \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Pivot columns $1, 2$ leave x_3 free, with $x_1 = x_2 = -x_3$. Hence the null space is $\{t(-1, -1, 1)^T : t \in \mathbb{R}\}$. The two nonzero rows have distinct leading positions and are independent. The nonpivot column is the sum of the pivot columns.

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Elimination operations and intermediate states (continued)

M=BC: dimension2 from original pivot columns

Original M:3×3,rank2

1	2	3
2	4	6
0	1	1

=

B: original pivot columns

1	2
2	4
0	1

×

C: column coordinates

1	0	1
0	1	1

M.column3=column1+column2, so C.column3=$(1,1)^T$.

B retains original vectors. Reduced unit columns give coordinates and cannot replace B.

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Gauss–Jordan without assuming diagonal pivots

```
R = copy(A[m,n]); r = 0; pivots = empty list
for j=1..n
    find t in r+1..m with R[t,j] != 0
    if none: continue
    r = r+1; swap rows r,t
    divide row r by R[r,j]
    for i=1..m with i != r
        a = R[i,j]
        row i = row i - a * row r
    append j to pivots
return R, pivots
```

Before each step, the first r pivot columns are distinct unit columns, and unused rows are zero in those columns. A new pivot comes from a remaining row. Normalizing it and eliminating elsewhere preserves old pivot columns because its old pivot coordinates are zero. Skip a column only when every remaining row is zero there. The finite loops produce r pivots and zero rows in $O(mn \min(m,n))$ field operations.

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Gauss–Jordan without assuming diagonal pivots (continued)

Exact comparison with zero follows the field arithmetic. Floating-point arithmetic needs pivoting, scale, and tolerance analysis. A fixed rule such as $|\text{pivot}| < 10^{-12}$ is not a universally correct mathematical definition of rank.

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Why row rank equals column rank

Let $R = EA$ be the reduced row echelon form, with E invertible. Applying the same reversible map to every column preserves exactly their dependence coefficients. The r pivot columns of R are independent, and every other column is their combination. Thus the column rank of A is r.

Invertible row operations preserve row span. The r nonzero rows of R are independent: their successive pivot coordinates force each coefficient in a dependence relation to vanish. Thus the row rank of A is also r. Write $\operatorname{rank}(A)$ for this common value, between 0 and $\min(m, n)$.

The nullity is $n - r$. Free coordinates uniquely determine the pivot coordinates. Set one free variable at a time to 1 and the others to 0 to construct a null-space basis. These vectors are independent by their free coordinates and span every solution. This proves the dimension statement generally, beyond the example M.

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Rank factorization and product bounds

Choose the r original pivot columns of A to form $B \in \mathbb{R}^{m \times r}$. Put the coordinates of every original column relative to B in $C \in \mathbb{R}^{r \times n}$. Then $A = BC$, the columns of B are independent, and C contains identity columns. Conversely, if $A = UV$ with s intermediate columns, every column of A lies in the span of U, so $\operatorname{rank}(A) \leq s$. Therefore r is the minimum possible intermediate dimension.

For M,

$$B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad M = BC.$$

The pivot vectors of R are not the original vectors of M; row operations changed their entries.

For compatible A, D, factor $A = B_r C_r$ to get $AD = B_r(C_r D)$, so $\operatorname{rank}(AD) \leq \operatorname{rank}(A)$. Factoring $D = U_s V_s$ similarly gives $AD = (AU_s)V_s$ and the bound by $\operatorname{rank}(D)$. Thus

$$\operatorname{rank}(AD) \leq \min\{\operatorname{rank}(A), \operatorname{rank}(D)\}.$$

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Rank factorization and product bounds (continued)

For an invertible left factor P, the identity $A = P^{-1}(PA)$ proves the reverse inequality, giving $\operatorname{rank}(PA) = \operatorname{rank}(A)$. Invertible right multiplication is analogous.

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Inverse existence and uniqueness

An inverse B of a square $n \times n$ matrix A satisfies $AB = I_n = BA$. If B, C are both inverses, $B = B(AC) = (BA)C = C$. For invertible A, B, the product inverse is $(AB)^{-1} = B^{-1}A^{-1}$, as multiplication on both sides verifies. Transposing the two inverse equations gives $(A^{-1})^T = (A^T)^{-1}$.

Square full rank is equivalent to no nonzero null vector, hence to injectivity of $x \mapsto Ax$. The n independent columns form a basis of n-space. Each e_j therefore has a unique solution $Ab_j = e_j$. Assemble these columns into B to obtain $AB = I_n$. Then $A(BA - I_n) = 0$ and injectivity imply $BA - I_n = 0$ columnwise. Thus full rank is equivalent to invertibility. Conversely, multiplying $Ax = 0$ by the inverse gives $x = 0$.

A rectangular one-sided inverse need not be two-sided. For $A = \begin{bmatrix} 1 & 0 \end{bmatrix}$ and $B = (1, 0)^T$, one has $AB = I_1$ but $BA = \operatorname{diag}(1, 0) \neq I_2$. The square premise cannot be dropped.

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A fully checkable inverse trace

Let $H = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$. Apply each row operation to both halves of $[H \mid I_2]$:

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - (3/2)R_1} \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1/2 & -3/2 & 1 \end{array}\right].$$

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1/2 & -3/2 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow 2R_2} \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1 & -3 & 2 \end{array}\right].$$

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Finish and verify the inverse

Subtract row two from row one, then normalize row one:

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1 & -3 & 2 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{cc|cc} 2 & 0 & 4 & -2 \\ 0 & 1 & -3 & 2 \end{array}\right].$$

$$\left[\begin{array}{cc|cc} 2 & 0 & 4 & -2 \\ 0 & 1 & -3 & 2 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1/2} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \end{array}\right].$$

The right half is $H^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$. Both checks give the identity:

$$HH^{-1} = \begin{bmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{bmatrix} = I_2, \qquad H^{-1}H = \begin{bmatrix} 4-3 & 2-2 \\ -6+6 & -3+4 \end{bmatrix} = I_2.$$

Elimination applies E to the left half while turning the right half into E. When $EH = I_2$, E is the inverse. Without all n left-side pivots for an $n \times n$ matrix, the right half cannot simply be declared an inverse.

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Determinants, minors, and cofactors

For a square matrix A, the minor matrix $A[ij]$ removes row i and column j. The cofactor is $C_{ij} = (-1)^{i+j} \det A[ij]$. For $n = 1$, define $\det[a] = a$. The optional 0×0 convention $\det(\varnothing) = 1$ unifies cofactor formulas. Expansion along row i gives $\det A = \sum_{j=1}^{n} a_{ij} C_{ij}$.

For $N = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & -1 & 1 \end{bmatrix}$, first-row signs $+, -, +$ give

$$\det N = 2\big(3 \cdot 1 - 2(-1)\big) - 1(1 \cdot 1 - 2 \cdot 0) + 0 = 10 - 1 = 9.$$

A minor is a matrix, its determinant is a scalar, and a cofactor adds a sign. These are distinct objects.

In general, with S_n the permutations of $\{1, \dots, n\}$,

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Each term selects one entry per row and per column, with a permutation-parity sign. Grouping terms by the selected column in a fixed row gives cofactor expansion.

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Reasons for determinant laws

The permutation formula is linear separately in every column. With two equal columns, pair permutations that exchange those columns; the terms cancel, giving determinant zero. Swapping columns changes the sign. The same arguments apply to rows, and the bijection $\sigma \mapsto \sigma^{-1}$ gives $\det(A^T) = \det A$. Scaling a row by α scales the determinant by α. Adding a multiple of another row leaves it unchanged because the extra term from multilinearity has a repeated row.

For any alternating multilinear scalar function F of the columns, expand each column in the standard basis. Repeated basis indices vanish. The remaining permutation terms satisfy

$$F(e_{\sigma(1)}, \ldots, e_{\sigma(n)}) = \operatorname{sgn}(\sigma) F(I_n).$$

Thus $F(B) = F(I_n) \det B$. Fix A and take $F(B) = \det(AB)$. Each product column is A times a column of B, so F is alternating multilinear and $F(I_n) = \det A$. This proves $\det(AB) = \det A \det B$ without assuming A invertible. If $\det A = 0$, the function is identically zero.

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Determinants and singularity beyond the 2×2 formula

For an upper-triangular matrix, a nonzero permutation term requires $\sigma(i) \geq i$ for every i. Since $\sum_i \sigma(i) = \sum_i i$, every inequality is an equality. Only the diagonal term remains, so the determinant is the diagonal product. The lower-triangular case follows symmetrically or by transpose. A square echelon matrix with a zero row has determinant zero; with all n pivots nonzero, it has a nonzero diagonal product.

Swaps and nonzero scalings multiply the determinant by nonzero factors; row addition preserves it. Therefore

$$\begin{aligned} \det A = 0 &\iff \text{a pivot is missing} \\ &\iff \operatorname{rank}(A) < n \\ &\iff A \text{ is singular.} \end{aligned}$$

A nonzero determinant proves invertibility, without guaranteeing good numerical conditioning. For $A = \operatorname{diag}(1, \varepsilon)$ with $\varepsilon \neq 0$, exact full rank holds but the inverse contains $1/\varepsilon$. Small ε can amplify errors. Stable factorizations usually solve $Ax = b$ without explicitly forming the inverse. Mathematical existence and numerical reliability are different issues.

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Real symmetric positive definiteness and Gram matrices

Here positive definite means a **real symmetric** matrix S with $x^T S x > 0$ for every nonzero real x. Positive semidefinite replaces $>$ with $\geq$. Without symmetry, a quadratic form sees only $(S + S^T)/2$; positivity alone does not imply $S = S^T$.

For real $A \in \mathbb{R}^{m \times n}$, the Gram matrix $G = A^T A$ is symmetric and

$$x^T G x = (Ax)^T (Ax) = \|Ax\|_2^2 \geq 0.$$

Equality holds exactly when $Ax = 0$. Thus G is positive definite exactly when A has no nonzero null vector, or equivalently has full column rank. This is impossible when $n > m$. A merely nonzero A is insufficient.

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Check Gram positivity on all vectors

For

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad X^T X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

the vector $x = (s, t)^T$ gives $x^T X^T X x = s^2 + t^2 + (s + t)^2 > 0$ unless $s = t = 0$.

For

$$W = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}, \quad W^T W = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix},$$

the quadratic form is $5(s + 2t)^2$. The nonzero vector $(-2, 1)^T$ has $Wx = 0$, so this Gram matrix is only positive semidefinite.

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Check Gram positivity on all vectors (continued)

Compare $x^TA^TAx=\text{norm}(Ax)^2$ at two ranks

X : full column rank



W : dependent columns



Same nonzero $x=(-2,1)^T$. Positive definiteness needs all nonzero x; one null vector refutes W.

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Complex and finite fields change some premises

Field arguments for multiplication, inverses, rank, and determinants extend to $\mathbb{C}$ and prime fields $\mathbb{F}_p$. Real order and Euclidean inner products do not transfer unchanged. Arithmetic modulo a composite number forms a ring in which nonzero elements may be noninvertible. For example, 2 modulo 6 has no inverse, so pivot normalization and field-rank arguments require reconsideration.

The complex conjugate transpose is A^*. A Hermitian matrix satisfies $S^* = S$. The Gram matrix A^*A is positive semidefinite and is positive definite for full column rank because

$$x^*A^*Ax = \sum_i |(Ax)_i|^2.$$

Ordinary A^TA need not be positive. For $A = [i]$, one has $A^TA = [-1]$ but $A^*A = [1]$.

For real or complex square A, the algebraic transpose identity is $\det(A^TA) = (\det A)^2$. The conjugate version is $\det(A^*A) = |\det A|^2 \geq 0$. The first expression implies nonnegativity only for real entries. Distinguish algebraic transposition from conjugation in inner products.

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Sources

Bibliography

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References and attribution

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